Effect of providers' characteristics on total served beneficiary: Analysis of 2020 Medicare physician's and practitioners data

Pranta Das

University of Nebraska-Lincoln

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Outline

1. Introduction to the dataset
2. Vision behind the research
3. Data quality checking and cleaning process
4. Exploration of the dataset
5. Modelling
6. Recommendations
7. Limitations

Introduction

The 2020 Medicare Physician & Other Practitioner by Provider and Service data.

- Collected by Center for Medicare & Medicaid Services(CMS)
- Updated annually.
- Data is available from 2013 to 2020.
- Contains information
  - Charges
  - Total beneficiary
  - Total Services
- Organized by
  - National Provider Identifier(NPI)
  - Service code
  - Place of service

Vision behind the research

Information regarding following variables were presented in the dataset.

Variable	**Description**
National Provider Identifier (**NPI**)	NPI is the numeric identifier registered
HCPCS Code	This code identifies the specific medical service furnished by the provider.
Entity type of the provider	This tells us whether the provider registered as an **Individual (I)** or **Organization (O)**
Medicare Participation Indicator(**MPI**)	This tells us whether provider participates in Medicare assignment or not.
Place of Service	This tells whether the service is given in **facility(F)** or **non-facility(O)**.
Gender	Represents gender.
Number of services (**Total Services**)	Number of services provided.
Number of Medicare Beneficiary (**Total Beneficiary**)	Number of Distinct Medicare beneficiary receiving the service

Checking the quality of data

Used 2019 data and employed following steps

1. Took out all the common NPI's in 2019 and 2020.
   - Sample size of 2020 is now 8.5M+
   - Sample size of 2019 is now 9.1M+
2. Merged two years data using variable NPI and HCPCS Cd.
   - Resulted in rows having data of 2019 and 2020 for each NPI and HCPCS Cd.
   - Narrowed down sample size to 6.6M+
3. Compared variable values in 2020 with values in 2019 and started cleaning anomalies.

Cleaning regarding gender

Following table represents

Table: Comparison of variable gender in both years.

		2020		
	Gender	**M**	**F**	**O**
2019	**M**	4,382,267	132	0
	F	92	1,969,763	0
	O	0	0	2,84,502

Omitting those rows reduced the sample size to 6,636,612 i.e 6.6M+

Cleaning regarding total beneficiary

- Total beneficiary in 2020 should be similar to total beneficiary in 2019 for a provider.
- So scatterplot will result in roughly a straight line

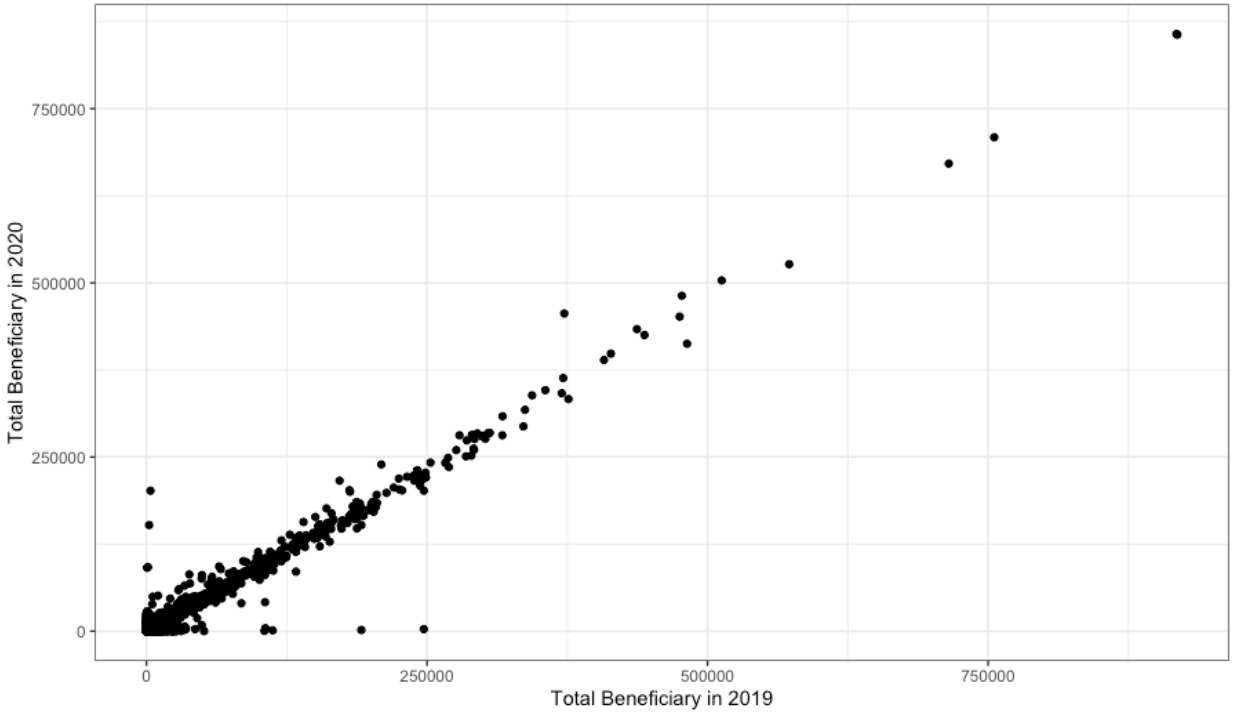

Figure: Scatterplot of total beneficiary in 2020 vs. in 2019.

Cleaning process regarding Total Beneficiary

Employed following steps

1. Standardized total beneficiary for respective years using

$$\frac{TotalBeneficiary - median(TotalBeneficiary)}{IQR(TotalBeneficiary)}$$

2. Took absolute difference of standardized values between 2020 and 2019.

$$Diff = |STB_{2020} - STB_{2019}|$$

3. Diff $\geq 6 \rightarrow$ unusual and 54,010 of them were found.

Omitting them, narrowed down the sample size to 6,582,602 which is 6.5M+

Comparison of Total Beneficiary between 2019 and 2020

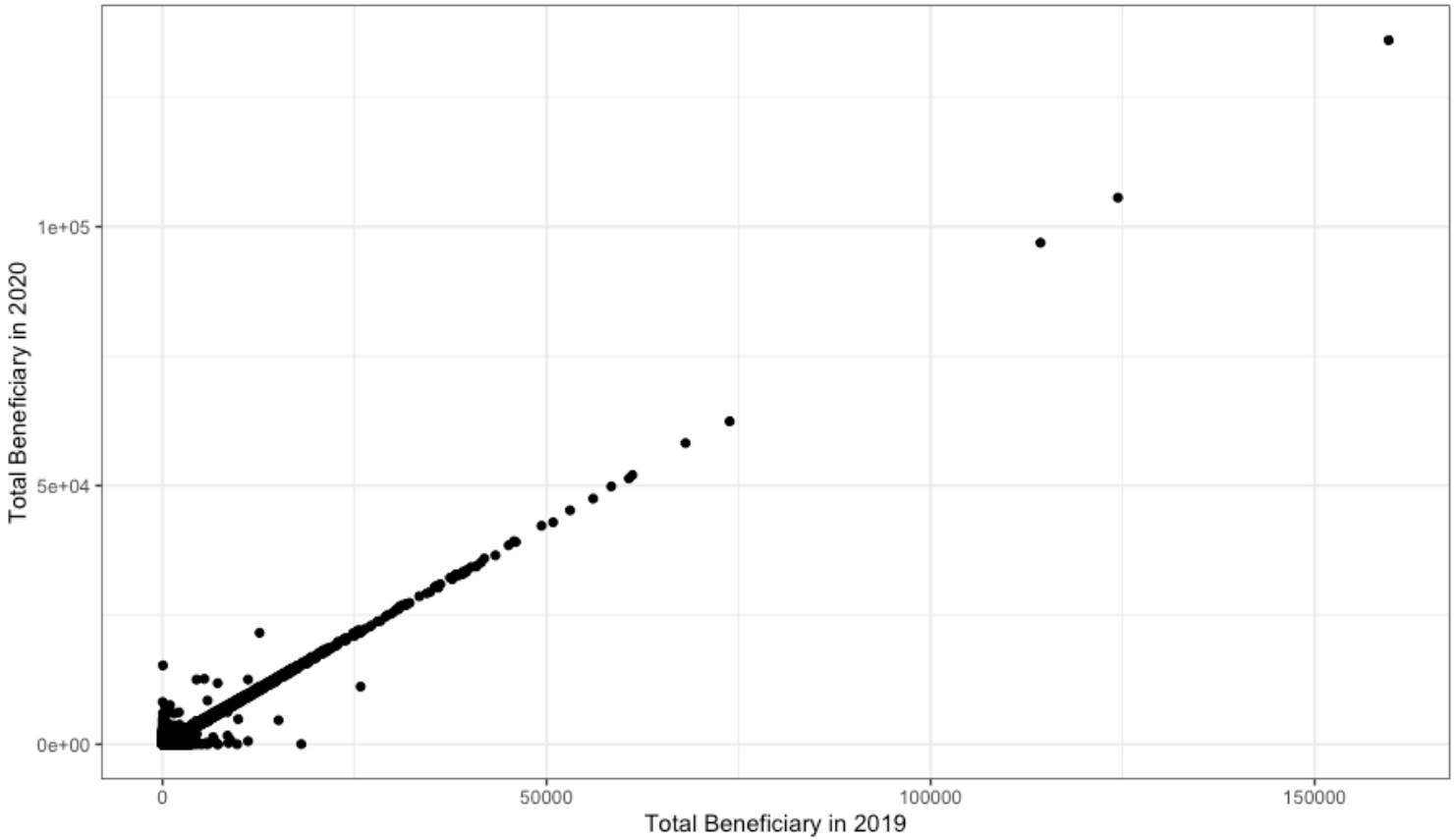

Figure: Scatterplot of total beneficiary in 2020 vs. in 2019 based on cleaned data.

Cleaning process regarding Total Services

- Same as I did for Total Beneficiary
- Omitted 72,902 rows and narrowed sample size to 6,509,700 i.e 6.5M+

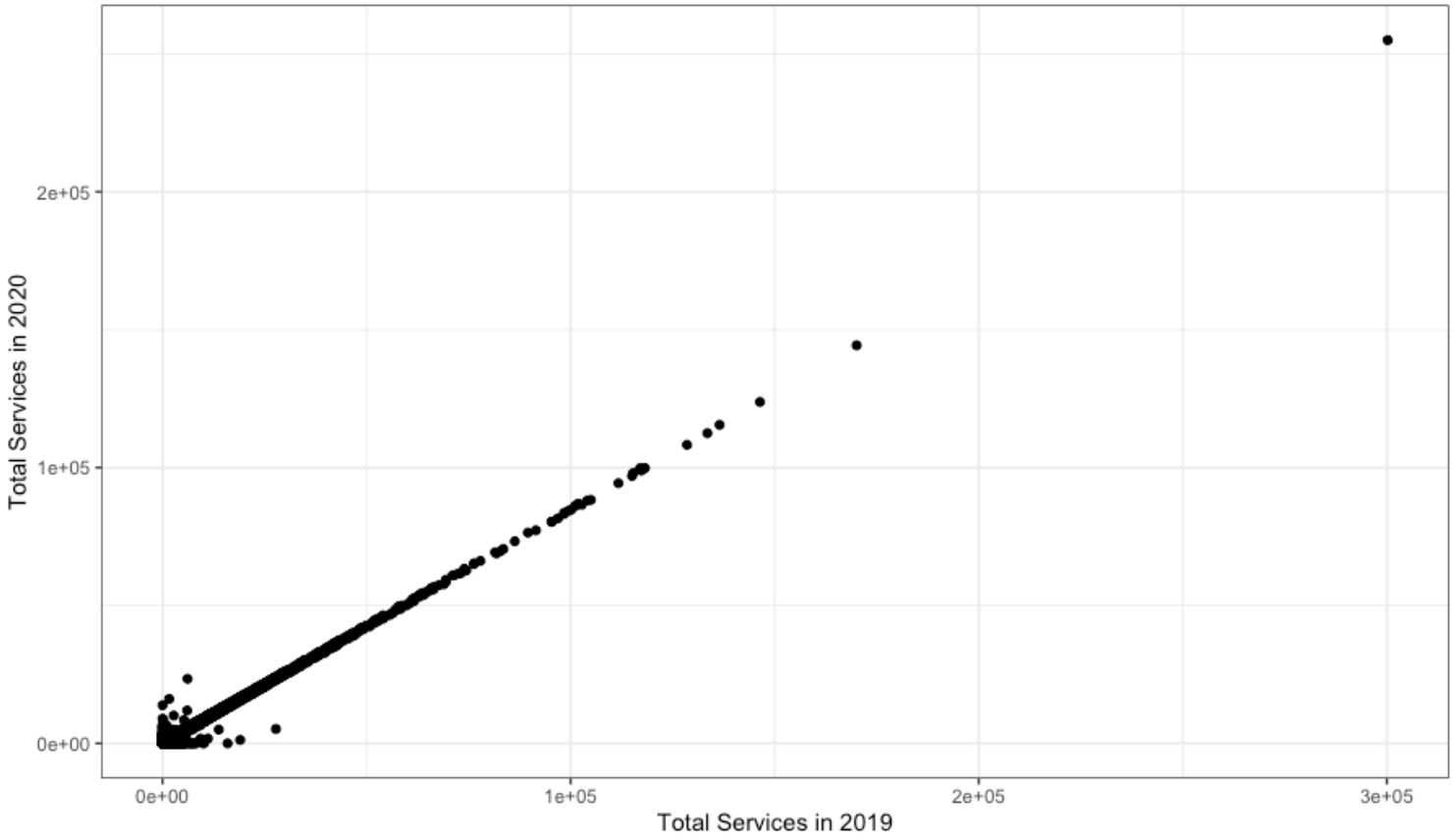

Figure: Scatterplot of total services in 2020 vs. in 2019 based on cleaned data.

Total Beneficiary and providers' entity

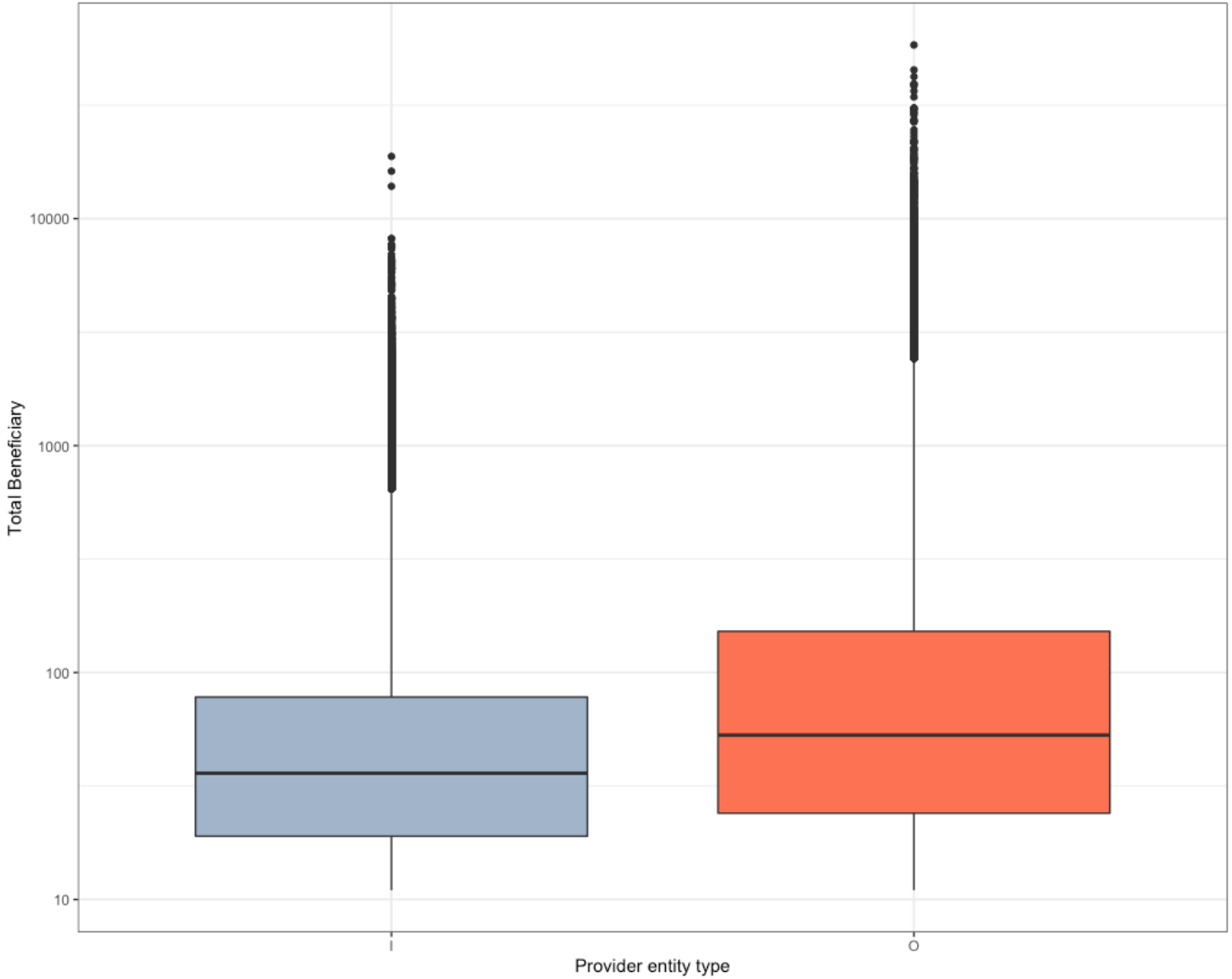

Figure: Side-by-side boxplot of total beneficiary with respect to provider entity.

Total Beneficiary and place of service

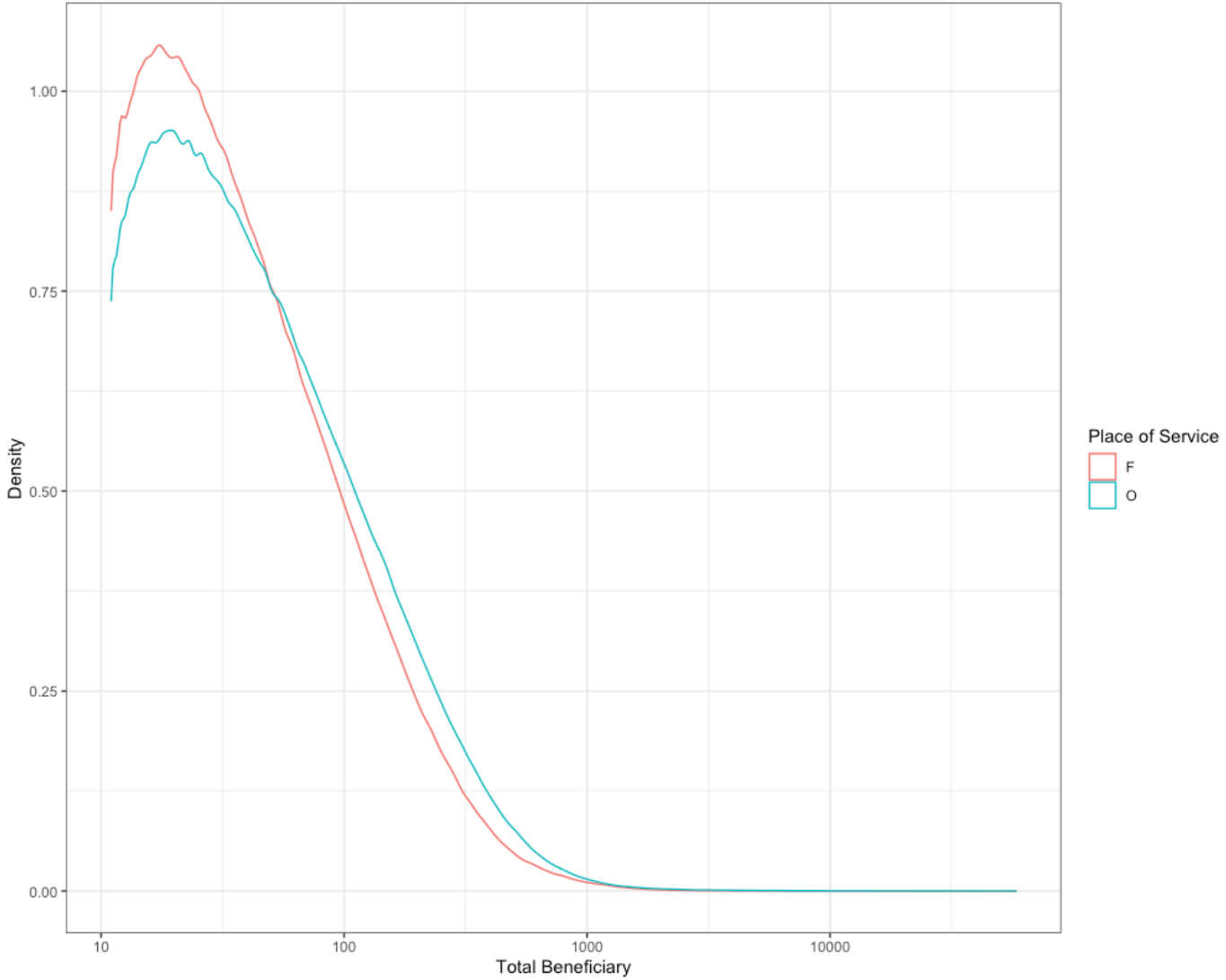

Figure: Density plot of total beneficiary with respect to place of service.

Total Beneficiary and medicare participation

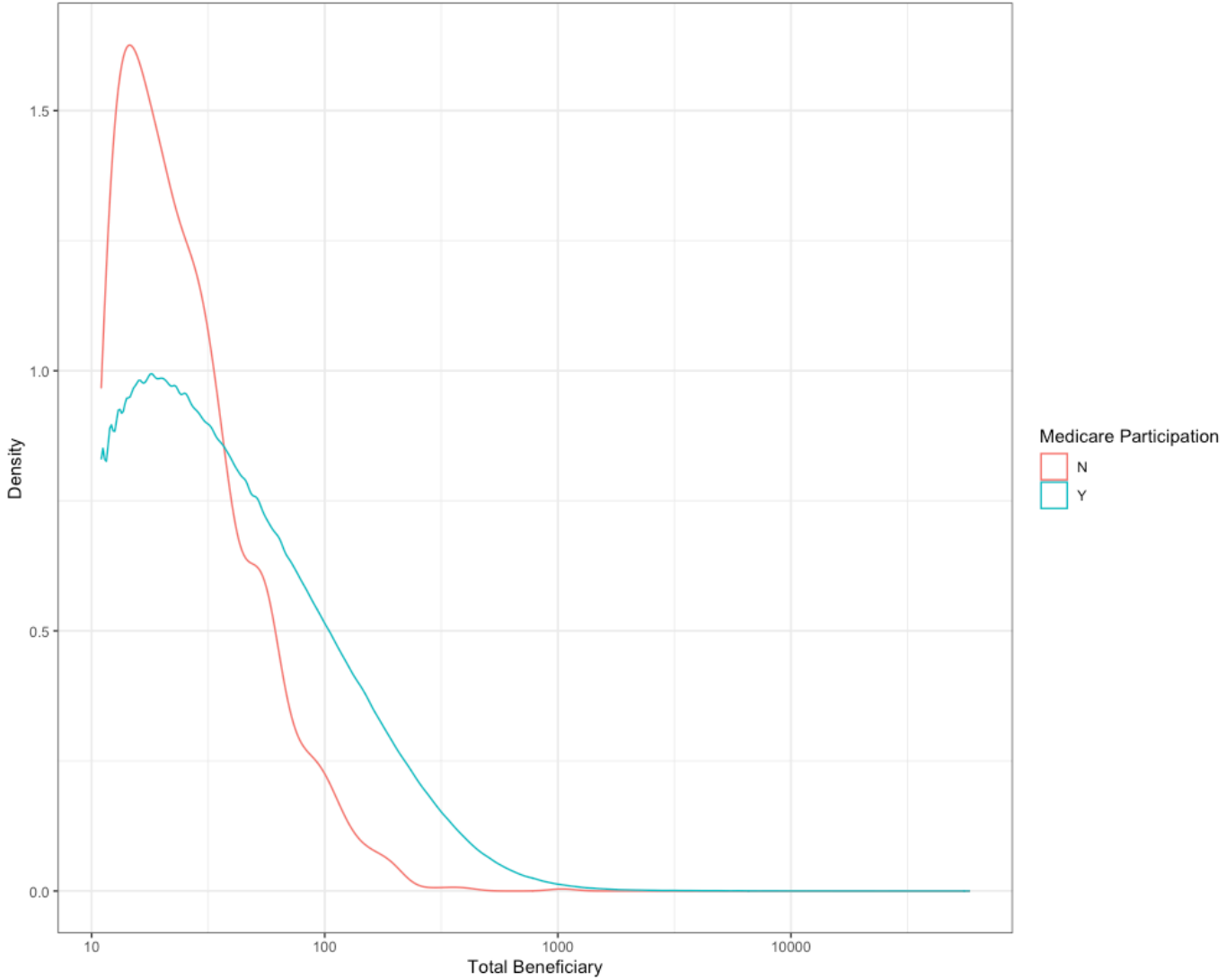

Figure: Density plot of total beneficiary with respect to medicare participation.

Total Beneficiary and Total services

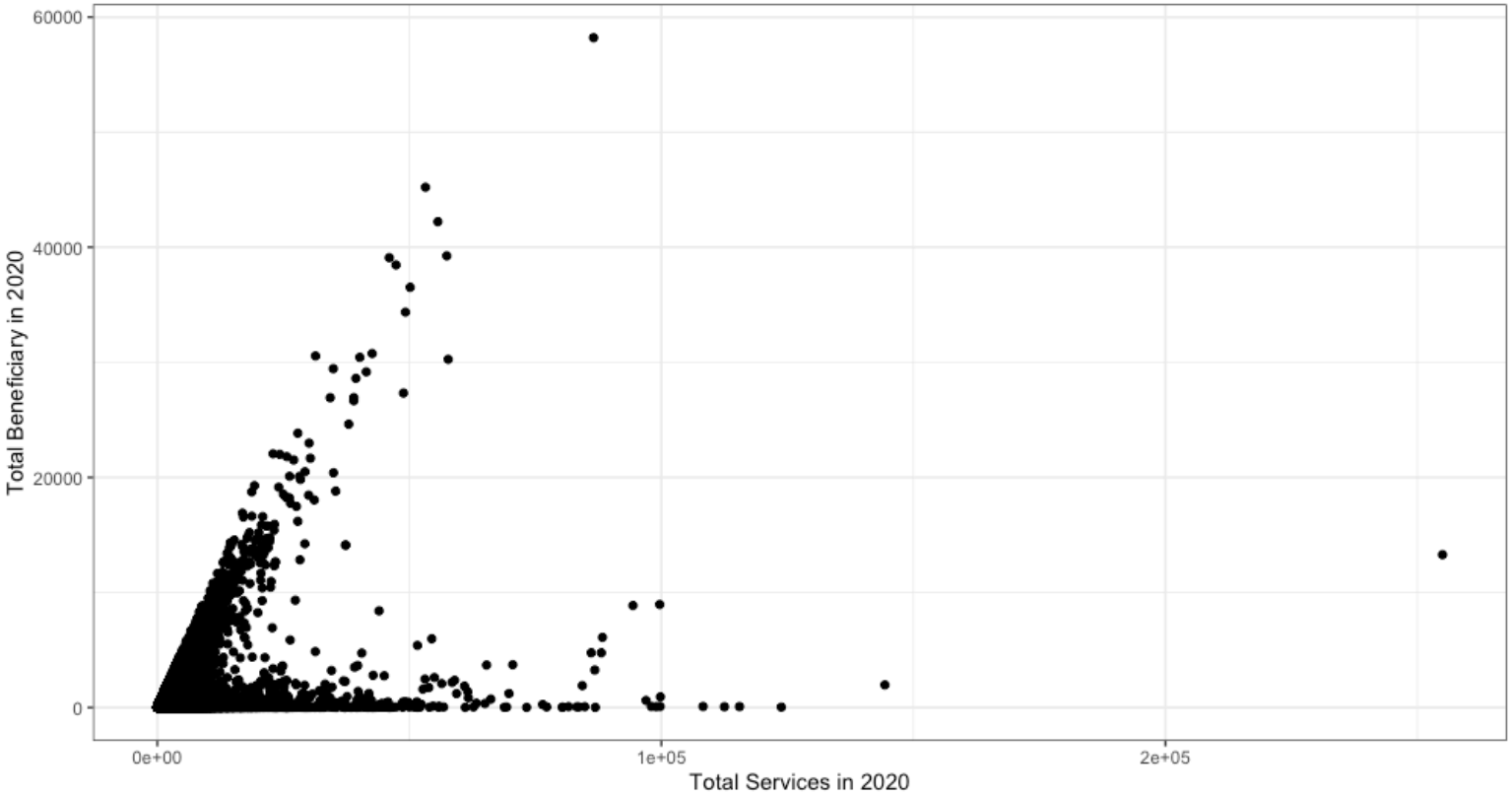

Figure: Scatterplot of total beneficiary in 2020 vs. total services in 2020.

Comment: Changes are visible in total beneficiary.

Conceptual model

What I observed?

- Changes in total beneficiary
  - Entity type
  - Place of service
  - Medicare participation
  - Total Services
- Changes but don't know how much.

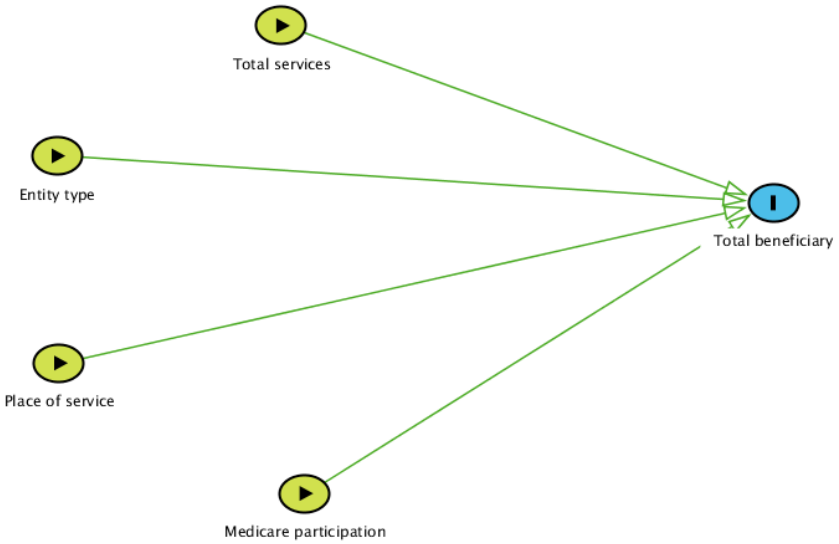

Figure: Conceptual model.

Nature of response variable

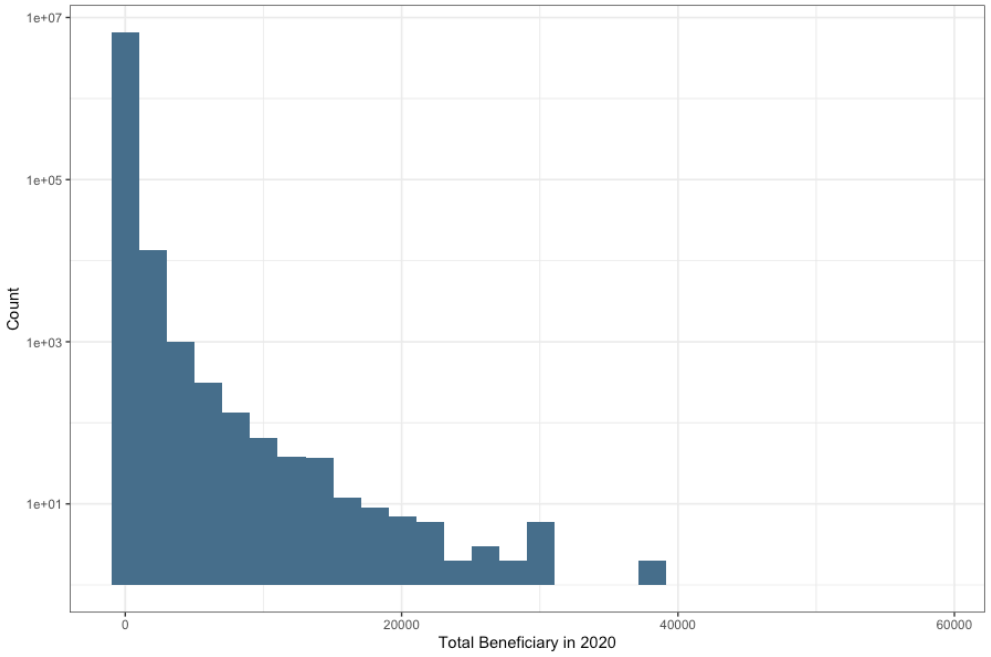

Figure: Histogram of Total Beneficiary

- Count
- Non-normal distribution
- Possible models
  - Poisson
  - Negative Binomial

Final Model

Table: Results obtained from Negative Binomial model.

Variable	Mean difference	95% CI	p-value
Place of Service			
Non-facility (O) - Facility (F)	1.164	1.162 - 1.166	<0.001
Providers' entity			
Organization (O) - Individual (I)	2.458	2.449 - 2.468	<0.001
Medicare Participation			
Yes - No	2.275	2.173 - 2.381	<0.001

Possible things to do based on this work

To the providers, if possible

- Register as an Organization rather than Individual.
- Change service place to non-facility.
- Participate in medicare assignments.

Limitations of my work

- Not a random sample according to CMS.
- Didn't consider other possible variables like Avg. cost submitted, State, etc.
- Didn't consider new providers'.
- Observational in nature.

Thank you.

(pdas8@huskers.unl.edu)